Calibration-Aware Test-Time Adaptation for CLIP

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Abstract

Test-time adaptation (TTA) methods such as TDA [3] use only forward passes and a feature cache to recover much of the accuracy CLIP loses out of distribution, but a 2025 benchmark [5] showed that the gain comes with a hidden cost: TDA's cache rewards confident predictions whether or not they are correct, so its accuracy gain arrives bundled with worse calibration (the benchmark reports average ECE rising 5.70% → 9.21% across its suite). We propose three training-free, label-free changes to TDA's cache loop – margin-based cache admission, probabilistic fusion, and leave-one-out (LOO) temperature scaling – and measure each one on five datasets (`ViT-B/16`, test split only) against a TDA-equivalent baseline. Margin admission fails and is dropped: its fixed threshold starves the cache on hard datasets and biases it toward easy classes. Fusion fails as proposed, because a nearly flat cache distribution averaged into CLIP's peaked one caps every prediction's confidence near one half; but *gating* the fusion to samples where CLIP is torn between two classes, and leaving the cache out elsewhere, repairs it. Gated fusion alone lowers mean ECE from TDA's 11.24% to 4.61%, below zero-shot CLIP's own 5.57%, at a mean accuracy of 55.01% against TDA's 56.63% and CLIP's 53.16%. LOO temperature scaling helps on its own (mean ECE 8.54% at unchanged accuracy) but does not stack on the gate, since both remove the same overconfidence. The method we recommend is TDA's cache with gated fusion: one extra comparison per sample, no labels, no training.

1 Introduction

CLIP's zero-shot accuracy is strong on natural-image classification but collapses on out-of-distribution domains: the benchmark that motivates this project reports 23.7% on EuroSAT and 15.7% on Aircraft [5]. Our own zero-shot measurements (Table 6) are higher – 42.06% and 23.85% – because we evaluate the CoOp test splits with a `ViT-B/16` backbone rather than that paper's exact setup; the collapse relative to in-distribution accuracy is the point, and it reproduces. Test-time adaptation (TTA) methods close much of this gap without any labels or backpropagation. TDA [3], in particular, keeps a small cache of features from test images the model is confident about and reuses that cache to refine later predictions – a forward-pass-only method with no training loop.

Accuracy, however, is not the only axis that matters for a system meant to be trusted. A model that is right 80% of the time but reports 95% confidence on every prediction is *worse* to deploy than one that is honest about its 80%, because a downstream user or pipeline has no way to tell a confident-and-correct prediction from a confident-and-wrong one. This is exactly the failure mode Sheng et al. [5] document in TDA: its cache rewards confident predictions with more confidence regardless of correctness, so accuracy gains arrive bundled with a calibration loss (average ECE rising 5.70% → 9.21% across their benchmark).

This project asks whether that calibration gap can be closed without retraining and without labels, by changing three seams of TDA's cache loop:

1. **Margin-based cache admission.** Admit a test sample to the cache by the gap between its top-1 and top-2 class scores, instead of by prediction entropy, so the cache does not accumulate merely *loud* predictions.

2. **Probabilistic fusion.** Combine CLIP's and the cache's evidence as a convex mixture of probability distributions, instead of adding raw logits, so agreement between the two sources cannot manufacture confidence that neither source had on its own.

3. **Leave-one-out (LOO) temperature scaling.** Estimate the cache's own accuracy without ground truth, via leave-one-out nearest-neighbour agreement among cached items, and pick a softmax temperature that makes the model's mean confidence match that label-free estimate.

Each change is independently switchable from configuration, so each can be measured on its own against a TDA-equivalent baseline, and Section 5 does exactly that, in order. The first change fails and is dropped. The second fails as proposed, but the diagnosis of why points at a repair – apply the fusion only where CLIP is torn between two classes – and the repaired version is the method this report recommends: it brings TDA's calibration error below zero-shot CLIP's while keeping most of TDA's accuracy gain. The third helps on its own but adds nothing on top of the second. We report each failure alongside its mechanism rather than only the configuration that worked.

2 Related Work

Training-free test-time adaptation for CLIP. TDA [3] adapts CLIP at test time with two small feature caches – a positive cache keyed by the model's own predicted class, and a negative cache of low-confidence samples used as a correction term – built and consumed purely from forward passes, with no backpropagation and no labels. Each new test image is scored by CLIP and optionally admitted to the positive cache by an entropy threshold; the final logits are a sum of CLIP's own logits and an α/β-weighted cache-logit term computed from the currently cached features. This project's method subclasses TDA directly and reuses its cache-update and cache-logit machinery unmodified; the three contributions in Section 3 change only the admission rule, the CLIP/cache combination rule, and a final temperature.

The calibration cost of TTA. Sheng et al. [5] benchmark a broad set of TTA methods for vision-language models, including TDA, across the same class of few-shot recognition datasets used here, and report accuracy alongside expected calibration error (ECE). Their finding motivates this entire project: TTA's forward-pass-only adaptation loop tends to raise accuracy while *also* raising ECE, because methods that reward and reuse confident predictions have no mechanism to distinguish confident-and-correct from confident-and-wrong. Their reported average ECE for TDA rises 5.70% $\rightarrow$ 9.21% on their suite even as accuracy improves. This project targets that specific gap using TDA as the adapted method and their framing of the problem, while running fully independent experiments (Section 4) rather than reusing their reported numbers.

Temperature scaling and ECE. Expected calibration error (ECE) buckets predictions by confidence and measures the gap between confidence and empirical accuracy within each bucket, weighted by bucket occupancy [1]. Temperature scaling divides logits by a single learned scalar T before the softmax. The operation is rank-preserving (it cannot change which class a model predicts) but reshapes confidence, and T is normally fit by minimizing negative log-likelihood on a labelled validation set. This project cannot use a labelled validation set – the spec's no-ground-truth constraint applies throughout adaptation – so Section 3 replaces the usual supervised fit with a leave-one-out, label-free accuracy estimate computed from the method's own cache.

3 Method

All three changes are implemented as independently switchable options on top of upstream TDA's runner. With every option off, our method reduces exactly to upstream TDA – same caches, same per-dataset α/β hyperparameters, same additive combination – which is the equivalence this project treats as a correctness invariant (verified by unit tests and, at the level of a full run, by the *none* rows of Tables 2–5 and Table 7).

3.1 Contribution 1: margin-based cache admission

Upstream TDA offers every test sample to its positive cache; entropy is used only as an eviction key for samples already in the cache, not as an admission gate. This project replaces unconditional admission with a decisive-prediction gate. Given the logit vector z for a sample, let $z_{(1)}$ and $z_{(2)}$ be its two largest entries. The *margin* is

$$m = z_{(1)} - z_{(2)}, \tag{1}$$

and a sample is admitted to the positive cache only when m exceeds a fixed global threshold (set to 1.0). Rationale: entropy conflates "the model is decisive" with "the model is decisive about a class it has seen a lot of evidence for"; the top-1/top-2 gap is a more direct decisiveness signal and does not require normalizing by the number of classes.

3.2 Contribution 2: probabilistic fusion

Upstream TDA finalizes a prediction as $z_{\text{clip}} + \alpha\, z_{\text{cache}}$, a sum of raw logits, and only then applies a softmax. This project instead fuses in probability space: given the CLIP logits z_{clip} and the cache logits z_{cache}, and a fusion weight $w \in [0, 1]$,

$$p = (1 - w)\, \sigma(z_{\text{clip}}) + w\, \sigma(z_{\text{cache}}), \tag{2}$$

where $\sigma(\cdot)$ is the softmax. Every run in this report uses $w = 0.5$. A convex combination of two distributions is bounded by the more confident of its two inputs, so two sources that happen to agree cannot combine into a confidence neither source had – exactly the guarantee additive logit-summing lacks, since summing before the softmax lets confident-and-agreeing logits amplify each other.

3.3 Contribution 3: leave-one-out temperature scaling

Standard temperature scaling fits a scalar T against a labeled validation set. No labels are available here, so T is instead chosen to match a label-free accuracy proxy computed from the method's own positive cache. Let the cache hold M feature vectors together with the class each was cached under (the class the method itself predicted for that image, not a ground-truth label). For each cached item i, find its nearest neighbour among the other $M - 1$ cached items by cosine similarity and check whether the neighbour's pseudo-label agrees with item i's own. The leave-one-out accuracy estimate is the fraction that agree:

$$\hat{a} = \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}\left[y_{\text{nn}(i)} = y_i\right], \tag{3}$$

where $\text{nn}(i)$ is item i's leave-one-out nearest neighbour and y denotes the cache's pseudo-labels. A well-calibrated classifier's mean top-1 confidence should equal its accuracy, so the temperature is

chosen to make the two match: writing $\bar{c}(T)$ for the mean top-1 softmax confidence over the run's score history after dividing logits by T,

$$T^\star = \arg\min_T \left| \bar{c}(T) - \hat{a} \right|, \tag{4}$$

found by grid search rather than a closed form. The search grid is deliberately *smoothing-only*: its floor is $T = 1.0$ (the identity), so no candidate can sharpen a prediction, only soften it. This is a design constraint, not an oversight. Margin admission selectively caches high-margin, likely-correct samples, which biases $\hat{a}$ upward relative to the model's true accuracy on the full test stream. An unconstrained search would respond to that inflated target by sharpening predictions, making overconfidence worse rather than better. Setting the grid's floor at $T = 1.0$ means the search can only push confidence down or leave it alone, never manufacture more of it. (This rationale is the reason the floor is kept, not the reason it was first added: an initial single-dataset run without it showed exactly the failure mode described above, sharpening predictions and worsening ECE. The mechanism holds regardless of which reason came first.) Temperature scaling is a monotonic rescaling of the logits, so applying it never changes a prediction, only its reported confidence.

All three contributions are independently switchable from configuration, so all eight on/off combinations can be run without any code change, which is the space the sweeps in Section 5 cover.

3.4 Code and attribution

This project is a fork of `TomSheng21/tta-vlm` [4], which supplies the evaluation framework used throughout: the CLIP wrapper, the dataset loaders, the entry points, and the TDA and zero-shot CLIP baselines every number in this report is compared against. The cache-update and cache-logit machinery that our method inherits and reuses unmodified originates in `kdiaaa/tda` [2] (MIT-licensed) as ported into that fork. Our own contribution is the calibration-aware method itself and the calibration, temperature-search, evaluation, analysis, and reporting code around it. The full source – including the scripts that regenerate every number, table, and figure in this report from the raw run records – is at `https://github.com/macwave12/deep_project_calibration_aware_tta`.

4 Experiments

4.1 Datasets and protocol

All numbers in this report are measured on the *test split only* of five standard few-shot recognition datasets, using the CoOp split files so counts match the benchmark literature exactly. Table 1 lists them:

Dataset	Test images	Classes
DTD	1,692	47
Oxford Flowers 102	2,463	102
Oxford Pets	3,669	37
EuroSAT	8,100	10
FGVC Aircraft	3,333	100

Table 1: The five evaluation datasets: test-split size and class count (CoOp splits).

Every method – zero-shot CLIP, TDA, and ours – is evaluated one image at a time in a single online pass over the test split, with no repeated epochs and no access to any dataset's training or validation split. Ground-truth labels are held out of every adaptation step: the cache, the margin gate, the fusion weight, and the LOO temperature search all operate on the model's own predictions ("pseudo-labels"), never on a true label; labels are opened only once per run, by the final scorer, to compute accuracy and ECE. The backbone is `ViT-B/16` throughout, with batch size 1 and seed 0.

Calibration is measured with expected calibration error (ECE), computed with 20 equal-width bins by a single implementation that scores every method identically.

Three sweeps produce every number in Section 5. The headline sweep covers five methods on the five datasets (25 configurations): zero-shot CLIP, TDA, and three configurations of our method. A per-contribution sweep runs our method under every on/off combination of the three proposed changes (40 configurations); only its baseline, margin-only, fusion-only and temperature-only variants are used, since the other four involve margin admission, which Section 5.1 drops, or ungated fusion, which Section 5.2 replaces. The ablation of Section 5.4 covers both on/off combinations of the two surviving changes with margin admission off (20 configurations); its baseline and temperature-only rows are the per-contribution sweep's, and Table 6's EuroSAT Temp and Gate+Temp cells are the ablation's. Runs are deterministic – every configuration that was executed twice produced identical records – so where two sweeps call for the same configuration (same code path, seed and effective settings) the run is executed once and its record reused. The scripts that regenerate all three sweeps from scratch are in the repository.

5 Results

The three contributions were proposed together, but they are not reported together. Each is measured first on its own, against the same baseline, labelled *none* in the tables: our method with every change switched off, which is upstream TDA run through our code path (Section 3). Comparing a contribution to this baseline rather than to the separately run TDA rows isolates that one change, since the two runs differ by nothing else. Where a contribution fails, we say why, and where the diagnosis suggests a repair, we test the repair before moving on. Only then are the surviving pieces compared against zero-shot CLIP and TDA together (Section 5.4).

5.1 Contribution 1: margin-based cache admission

Table 2 compares margin admission alone – margin on, fusion off, temperature off – with the baseline, per dataset, together with the fraction of test samples the gate admitted to the positive cache.

Margin admission loses on every dataset and on both metrics. Mean accuracy drops from 56.60% to 53.67% and mean ECE rises from 11.28% to 18.06%. The per-dataset spread is the informative part. The threshold is one global constant, 1.0 on raw logits, deliberately not tuned per dataset, because fitting it to each test set's own margin distribution would be test-set tuning. Since CLIP's logits are 100× a cosine similarity, a gap of 1.0 asks for a top-1/top-2 cosine difference of only 0.01, a quantity with no dataset-independent meaning. The admission column shows what that costs: the gate admits 15.9% of samples on Aircraft and 81.9% on Pets, and the rate is ordered by zero-shot CLIP's accuracy on the dataset, not by its class count (Flowers102 has the most classes and the second-highest rate; EuroSAT has the fewest and the second-lowest). Where CLIP already separates the classes, margins are wide and the gate opens; where CLIP is lost, every class looks

Dataset	none acc.	margin acc.	none ECE	margin ECE	Admission rate
DTD	47.04	44.86	17.41	20.11	52.96
Flowers102	69.79	67.19	6.51	8.63	65.69
Pets	88.74	88.42	2.48	2.53	81.90
EuroSAT	53.22	48.91	14.43	26.38	26.54
Aircraft	24.21	18.96	15.56	32.64	15.93
Mean	56.60	53.67	11.28	18.06	48.60

Table 2: Margin admission against the TDA-equivalent baseline (*none*, all flags off, run in the same sweep): accuracy and ECE (%) per dataset, and the fraction of test samples the margin gate admitted to the cache.

like every other, margins collapse, and the gate closes. The damage is ordered the same way: Pets, at 81.9% admission, is a near no-op (ECE +0.05 points), while Aircraft, at 15.9%, doubles its ECE.

Three things went wrong, and the admission rate explains only the first.

1. **The threshold does not port across datasets.** A fixed logit gap means different things on 10 and 100 classes, so one constant is simultaneously too loose on Pets and far too strict on Aircraft and EuroSAT.

2. **A starving cache is not a smaller cache; it is a biased one.** If the cache were merely thinner, predictions would drift toward plain CLIP and calibration toward CLIP's own 5%. Instead ECE doubles. At 16% admission, many classes are likely to have no cache entry at all (we record the admission rate, not per-class coverage): only the classes CLIP finds easy clear the threshold, so the cache fills with those classes and little else. A new image's nearest cached neighbours can then only ever vote for an easy class, whatever the image actually shows. The cache becomes a prior pulling every prediction toward the handful of classes CLIP was already confident about – which flips correct predictions on the under-represented classes (the accuracy drop) and adds those votes confidently (the ECE rise). Pets, with 37 classes and 82% admission, keeps every class covered, and nothing happens.

3. **It filters on the wrong signal.** The diagnosis was that confidently *wrong* samples enter the cache. But a confidently wrong prediction has a large margin, exactly like a confidently right one. Margin and entropy both measure how loud CLIP is, and neither sees correctness, so the gate could never remove the samples it was designed to remove. It only removed near-ties – and upstream TDA's per-class eviction by entropy was already removing those wherever a class saw enough traffic. The gate was redundant where it was harmless and biasing where it bit.

We therefore drop margin admission. Every variant from here on has margin admission off, and the positive cache is filled by upstream TDA's own unconditional rule.

5.2 Contribution 2: probabilistic fusion

Table 3 compares probabilistic fusion as proposed in Section 3 – fusion on with $w = 0.5$, margin off, temperature off – with the baseline.

Fusion as proposed is the worst single change in the project: mean ECE rises from 11.28% to 24.54%, driven by Flowers102 (6.51% $\rightarrow$ 32.46%) and Pets (2.48% $\rightarrow$ 43.78%). Accuracy moves

Dataset	none acc.	fusion acc.	none ECE	fusion ECE
DTD	47.04	45.80	17.41	17.69
Flowers102	69.79	68.90	6.51	32.46
Pets	88.74	88.77	2.48	43.78
EuroSAT	53.22	50.12	14.43	19.62
Aircraft	24.21	24.42	15.56	9.16
Mean	56.60	55.60	11.28	24.54

Table 3: Probabilistic fusion as proposed (ungated, $w = 0.5$) against the TDA-equivalent baseline (*none*): accuracy and ECE (%) per dataset.

little (mean 56.60% $\rightarrow$ 55.60%), so the cache is still doing its job of choosing the class; what breaks is the confidence attached to the choice.

The mechanism is visible in Eq. (2) once the two inputs' scales are compared. The cache logits, computed exactly as in upstream TDA, are per class a sum of a few $\exp(-\beta(1 - \text{affinity}))$ terms over the cached items of that class. With at most three items per class and affinity a cosine in $[-1, 1]$, the largest cache logit is about 2 and most classes score near zero. Softmaxed over 100 classes, that is a nearly flat distribution whose top entry is around 7%. CLIP's logits, by contrast, carry its learned logit scale (≈ 100) and are sharply peaked, often above 90%. A 50/50 average of a peaked distribution and a flat one halves the peak. Fusion therefore caps top-1 probability near one half on every sample, right or wrong alike – it does not need the two sources to agree on a wrong answer; it produces the effect unconditionally. ECE penalises underconfidence exactly as it penalises overconfidence, so the cost scales with accuracy: on Pets the model is right 88.77% of the time and reports roughly 45% confidence, a gap of about 44 points, which is the 43.78% in the table. Aircraft is the one dataset where fusion's ECE looks like an improvement (15.56% $\rightarrow$ 9.16%), and it is not a correction: Aircraft's true accuracy (24%) happens to sit close to where the confidence ceiling landed, so the same ceiling that ruins Pets lands near the truth there by accident.

Gating the fusion. The diagnosis says fusion fails because it runs on every sample, including the large majority where CLIP already has a clear winner and the cache can only dilute it. On those samples the cache cannot change the argmax either way, so the right move is to leave it out entirely: additive TDA would only sharpen (the original overconfidence) and fusion would only dilute. The cache has a real job on exactly one kind of sample – a two-way tie, where it can break the tie, which is where TDA's accuracy gain comes from – and on a genuine tie a confidence near one half is roughly honest. We therefore gate the fusion. Let $p = \sigma(z_{\text{clip}})$ and let $p_{(1)} \geq p_{(2)}$ be its two largest entries. A sample is a *two-way tie* when

$$p_{(1)} - p_{(2)} < 0.2 \quad \text{and} \quad p_{(1)} + p_{(2)} > 0.6. \tag{5}$$

The first condition is the close call; the second is what "low entropy" buys: a sample spread thinly over many classes also has a small top-two gap, but there CLIP is confused rather than torn, and the cache's vote is noise. On a tie the output is Eq. (2) with $w = 0.5$; otherwise it is $\sigma(z_{\text{clip}})$, CLIP's own distribution (corrected by TDA's negative cache, which is untouched throughout), with the positive cache not consulted. Both thresholds are in probability space, so one global pair means the same thing on 10 and 100 classes – the portability margin admission lacked – and both were fixed a priori before any run, not tuned. The gate is a separate switch, off by default, so every run above is unaffected.

Dataset	none acc.	gate acc.	none ECE	gate ECE
DTD	47.04	45.69	17.41	6.24
Flowers102	69.79	69.02	6.51	4.08
Pets	88.74	88.77	2.48	5.68
EuroSAT	53.22	47.30	14.43	3.56
Aircraft	24.21	24.27	15.56	3.48
Mean	56.60	55.01	11.28	4.61

Table 4: Gated probabilistic fusion against the TDA-equivalent baseline (*none*): accuracy and ECE (%) per dataset.

Table 4 compares gated fusion – gate on, margin off, temperature off – with the same baseline.

The blow-up is gone. Flowers102 and Pets return to single-digit ECE, and gated fusion beats the baseline's ECE on four of five datasets, by ten or more points on three of them (DTD 17.41% $\rightarrow$ 6.24%, EuroSAT 14.43% $\rightarrow$ 3.56%, Aircraft 15.56% $\rightarrow$ 3.48%). Mean ECE falls from 11.28% to 4.61%, which is below zero-shot CLIP's own 5.57% (Section 5.4): the method is better calibrated than the frozen model it started from. Pets is the one loss (2.48% $\rightarrow$ 5.68%), on the dataset where TDA was already nearly perfectly calibrated. The accuracy side is the cost. Gated fusion matches or fractionally exceeds the baseline on Pets and Aircraft, gives up 0.77 points on Flowers102 and 1.35 on DTD, and 5.92 on EuroSAT, where it returns more than half of TDA's eleven-point gain. The likely reason, which we did not instrument, is that TDA's EuroSAT flips happen on samples with a moderate margin, pushed over by that dataset's strong cache weighting ($\alpha = 4$, $\beta = 8$); the gate, by construction, keeps the cache out of those.

The gate uses the same quantity margin admission used – the gap between CLIP's top two scores – and the contrast between the two is the clearest finding in this section. Margin admission applied it at storage time, to decide what the cache remembers, and threw information away permanently and unevenly across classes. The gate applies it at prediction time, to decide when the cache is consulted, and leaves every class represented. Same test, opposite purpose: admission kept the clear winners and discarded the close calls, while the gate leaves the clear winners alone and brings the cache in only on the close calls, which are the only samples where the cache can change anything but confidence.

5.3 Contribution 3: leave-one-out temperature scaling

Table 5 compares LOO temperature scaling alone – temperature on, margin off, fusion off – with the baseline.

Accuracy is identical on every dataset, to 0.00 points. This is not a coincidence but a structural guarantee: temperature scaling is a monotonic rescaling of the logits and cannot change an argmax, so any difference would itself be a bug. On calibration it is a real but not uniform improvement. Mean ECE falls from 11.28% to 8.54%, with large gains where the baseline was badly overconfident (DTD 17.41% $\rightarrow$ 4.68%, Aircraft 15.56% $\rightarrow$ 4.21%) and small ones on EuroSAT and Flowers102. On Pets it is a clear reversal: 2.48% $\rightarrow$ 15.54%.

The Pets failure has a simple cause. The search has no stopping rule. It always moves mean confidence toward the label-free accuracy estimate $\hat{a}$, whether or not the model was already honest, so where TDA was already calibrated the only thing it can do is damage. And $\hat{a}$ is biased low on fine-grained data: the estimate is leave-one-out nearest-neighbour agreement inside the cache, and on Pets a cached Birman's nearest cached neighbour is often a Ragdoll even when both pseudo-

Dataset	none acc.	temp acc.	none ECE	temp ECE
DTD	47.04	47.04	17.41	4.68
Flowers102	69.79	69.79	6.51	4.85
Pets	88.74	88.74	2.48	15.54
EuroSAT	53.22	53.22	14.43	13.43
Aircraft	24.21	24.21	15.56	4.21
Mean	56.60	56.60	11.28	8.54

Table 5: LOO temperature scaling against the TDA-equivalent baseline (*none*): accuracy and ECE (%) per dataset. Accuracy is identical by construction.

labels are correct, so the agreement rate sits well below the true accuracy. The search reads that as "accuracy is lower than confidence" and smooths a model that did not need it. The smoothing-only grid floor at $T = 1.0$ (Section 3) cannot help here; it only prevents the opposite mistake.

5.4 Putting it together

Table 6 compares the five headline methods on all five datasets: zero-shot CLIP (CLIP), TDA, and three variants of our method – gated fusion alone (Gate), LOO temperature alone (Temp), and the two together (Gate+Temp). Margin admission is off in all three.

	Accuracy					ECE				
	CLIP	TDA	Gate+Temp	Gate	Temp	CLIP	TDA	Gate+Temp	Gate	Temp
DTD	44.39	47.04	45.69	45.69	47.04	8.27	17.41	6.21	6.24	4.68
Flowers102	67.32	69.91	69.02	69.02	69.79	2.63	6.39	5.89	4.08	4.85
Pets	88.20	88.77	88.77	88.77	88.74	4.37	2.52	13.90	5.68	15.54
EuroSAT	42.06	53.17	47.30	47.30	53.22	7.16	14.38	3.56	3.56	13.43
Aircraft	23.85	24.27	24.27	24.27	24.21	5.44	15.51	3.37	3.48	4.21
Mean	53.16	56.63	55.01	55.01	56.60	5.57	11.24	6.59	4.61	8.54

Table 6: Top-1 accuracy and ECE (%, 20 bins) on the five test splits. CLIP is zero-shot CLIP; Gate, Temp and Gate+Temp are our method with gated fusion, LOO temperature scaling, or both (margin admission off in all three). Mean is the unweighted mean over the five datasets.

Averaged over the five datasets, zero-shot CLIP reaches 53.16% accuracy at 5.57% ECE and TDA 56.63% at 11.24%. Gated fusion alone reaches 55.01% at 4.61%: it keeps most of TDA's accuracy gain over CLIP (1.85 of 3.47 points) while reporting better-calibrated confidence than either. Figure 1 shows the same comparison as a scatter: the target is a point as far right as TDA and as far down as zero-shot CLIP, and the purple diamond sits below both on three of the five datasets.

Table 7 is the ablation over the two surviving contributions: both on/off combinations of gated fusion and LOO temperature, margin off, on all five datasets, 20 runs.

The two contributions do not stack. Adding temperature on top of the gate leaves accuracy unchanged, as it must, and makes mean ECE worse (4.61% → 6.59%). The reason is that both target the same thing. Temperature scaling was designed to undo the overconfidence that TDA's cache manufactures; the gate removes that overconfidence at the source, by never letting the cache sharpen a prediction CLIP had already made. What is left after the gate is CLIP's own calibration,

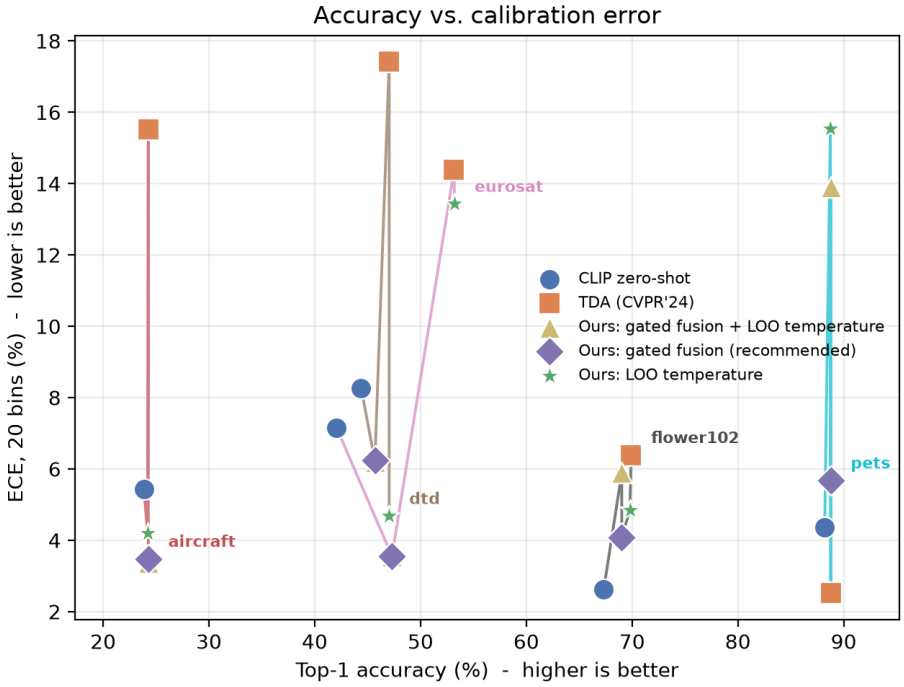


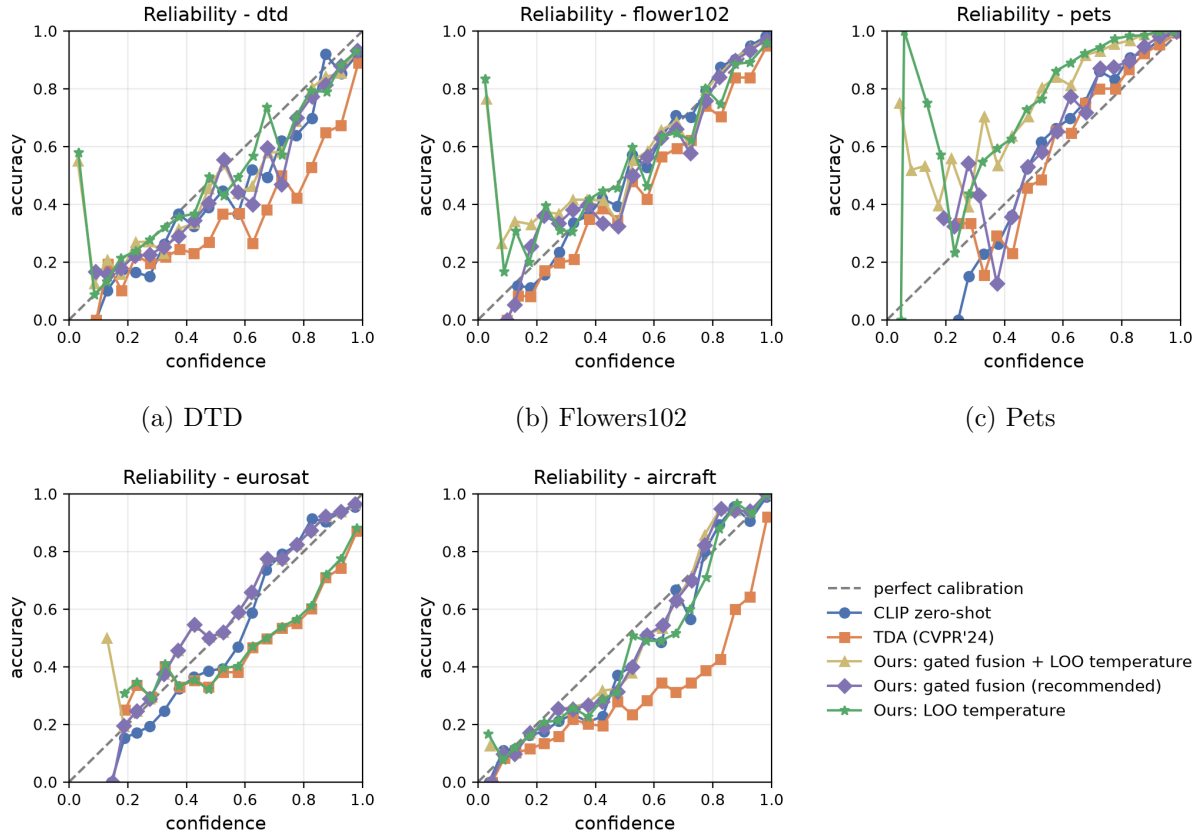
Figure 1: Accuracy vs. ECE on all five datasets for the five headline methods: zero-shot CLIP, TDA, and our method with gated fusion and LOO temperature together, gated fusion alone, and LOO temperature alone. Marker shape and colour identify the method (legend); each dataset's five points are joined by a line in that dataset's own colour, labelled at the line's right-hand end. Lines run left to right by accuracy and mark grouping only.

	Accuracy	ECE
none: TDA-equivalent (both off)	56.60	11.28
gate: + gated fusion	55.01	4.61
temp: + LOO temperature	56.60	8.54
gate_temp: + gated fusion + temperature	55.01	6.59

Table 7: Ablation with margin admission off: mean accuracy and ECE (%) across the five datasets for gated fusion and LOO temperature.

already around 5%, and the label-free estimate is not accurate enough to improve on it. On EuroSAT and Aircraft the search wants to sharpen, the floor at $T = 1$ blocks it, and the search is a no-op on 99.9% and 92.5% of samples respectively; on Pets it acts on every sample and in the wrong direction (5.68% $\rightarrow$ 13.90%), the same Pets failure as in Section 5.3. We therefore recommend gated fusion alone.

Figure 2 plots, per dataset, the reliability curve for each of the five headline methods. The direction of each failure is visible directly: TDA's curves sit below the identity line (stated confidence exceeds accuracy), the temperature-only curves sit above it on Pets (confidence below accuracy), and gated fusion tracks the diagonal most closely on DTD, EuroSAT and Aircraft.



(a) DTD (b) Flowers102 (c) Pets

(d) EuroSAT (e) Aircraft

Figure 2: Reliability diagrams, all five datasets, for the same five methods as Figure 1: zero-shot CLIP, TDA, gated fusion + LOO temperature, gated fusion alone, and LOO temperature alone. All five panels share the legend at bottom right; a curve on the dashed identity line is perfectly calibrated.

The success criterion, counted honestly. The project's criterion was accuracy $\geq$ TDA *and* ECE $<$ TDA on at least 3 of 5 datasets, against the TDA column of Table 6. Gated fusion meets it on Aircraft only: it ties TDA's accuracy on Pets but loses on ECE there, and on DTD, Flowers102 and EuroSAT it beats TDA's ECE by a wide margin while sitting below TDA's accuracy by 1.35, 0.89 and 5.87 points. Temperature alone meets it on 2 of 5 (DTD and EuroSAT; on the other three its accuracy is a few hundredths below the separately run TDA numbers, the floating-point drift

described in Section 5.5). No variant reaches 3 of 5. What the method does deliver is different from what the criterion asks for: it trades a small part of TDA's accuracy gain for calibration better than the frozen model's, and it does so without labels, without training, and with one extra comparison per sample.

5.5 Limitations

Every number here is one seed, one backbone (`ViT-B/16`) and five datasets. That is enough to establish that margin admission and ungated fusion fail, that the gate repairs fusion, and that temperature does not stack on it; it is not enough to claim the gate's margins hold universally, and the smaller effects (the one-point EuroSAT ECE change under temperature, the sub-point accuracy gaps on Pets and Aircraft) are within what a different stream order could plausibly move. The gate's two thresholds were fixed a priori and have been tested on nothing but these five datasets. Its accuracy cost on EuroSAT is real and would need a dataset-aware gap – which is to say, tuning – to recover. The LOO estimate is only as good as the cache it reads, and it is biased low on fine-grained data, which is what breaks temperature scaling on Pets. Finally, the baseline and the separately run TDA rows differ by 0–4 flipped predictions per dataset out of 1,692–8,100: upstream TDA accumulates its final logits in fp16 while our implementation upcasts to fp32 before its final step, and at CLIP's logit magnitudes that rounding step is worth about one unit in the last place, enough to flip a borderline argmax once every few hundred to few thousand samples. This is two orders of magnitude below any effect discussed above, but it is why the two baselines are two distinct, both-correct sets of numbers rather than duplicates.

6 Conclusion

This project set out to undo TDA's calibration loss with three training-free, label-free changes to its cache loop, and measured each one rather than assuming it worked. Margin-based cache admission fails outright: its fixed logit threshold does not port across datasets, a starved cache is a biased cache, and the top-two gap cannot tell a confident mistake from a confident success, so the gate removes the wrong samples. We dropped it. Probabilistic fusion fails as proposed, for a reason the equation makes visible once the two inputs' scales are compared: a nearly flat cache distribution averaged into CLIP's peaked one caps every prediction's confidence near one half, and ECE punishes that underconfidence exactly as it punishes overconfidence. The repair follows from the diagnosis. Fusing only where CLIP is torn between two classes, and leaving the cache out everywhere else, removes the dilution while keeping the one thing the cache is good for, breaking ties. Gated fusion lowers TDA's mean ECE from 11.24% to 4.61%, below zero-shot CLIP's own 5.57%, at a mean accuracy cost of 1.62 points, most of it on EuroSAT. Leave-one-out temperature scaling helps on its own, cutting mean ECE to 8.54% at unchanged accuracy, but reverses on the dataset that was already calibrated and adds nothing on top of the gate, because the gate has already removed the overconfidence it was built to correct.

The method we recommend is therefore TDA's cache with gated probabilistic fusion: one extra comparison per sample, no labels, no training, no change to the cache itself. Against the project's own success criterion it passes on one dataset of five, and we report that count as it stands. What the result supports is a narrower and, we think, more useful claim than the one we set out to make: that TDA's calibration loss comes from consulting the cache on samples where it can only add confidence, and that the gap between CLIP's top two scores, applied at prediction time rather than at admission time, is enough to tell those samples apart.

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